



Apple Music free trial slashed

Apple Music free trial slashed down from three months to one month in select countries.

<https://digit.in/mar22-53>



VR Boxing

Les Milles, a fitness club, is foraying into the metaverse with its new VR boxing game, Bodycombat VR. <https://digit.in/mar22-54>

particular location. The stiffness of the web is not altered much by the existing pretension. It does, however, serve as a key function in delaying the emergence of slack strings and so preventing nonlinearities and irregularities in symmetry caused by string slackening.

SPECIAL MENTION: THE ORB WEB

There are five known varieties of spider web: orb web, cobweb, funnel web, tube web, and sheet web. The orb web has been the most extensively researched of various sorts of webs. Its construction is a marvel of engineering requiring material tailoring, material interface optimisation, and resource conservation to ensure the spider's survival.

In addition, the web takes water from the atmosphere, and the spider's consumption of this water may contribute significantly to the spider's water intake demands. When a flying insect collides with the web, over 70 per cent of the energy is dissipated through viscoelastic processes. To prevent the web from breaking and the insect from being ejected from the web by elastic recoil, the web balances stiffness and strength against extensibility.

UNDERSTANDING THE SPIDERWEB

For the longest time, the method by which a spider spins its web has been under intense study, mostly due to its architectural accuracy. Everyone knows what the finished spider web looks like, but how does the spider start weaving it? Does it start from the centre, where the prey needs to be lured, or does it decorate the web from the periphery, slowly winding down to the point of focus? From a simple glance, we can see that there is a typical symmetry across the centre of a spider's web.

ALGORITHMIC SIMULATION USING MACHINE LEARNING

Gordus, a behavioural biologist at Johns Hopkins, and his colleagues built an arena with infrared lighting and cameras to film six spiders' web-building activities. A hackled orb weaver, native to the western United

States, was the subject of the study. They tracked the millions of individual leg movements the spiders made in order to create their webs using machine vision software designed specifically to detect limb movements. The scientists discovered that spider web-making behaviours were remarkably similar across species.

Web construction, like other examples of animal architecture, is the outcome of numerous assembly processes driven by different behavioural stages, coordinated to form a successful structure. Although manual observations have revealed a variety of sensory signals and movement patterns used during web formation, there are no ways for quantifying the dynamics of these sensorimotor patterns in a systematic way.

The scientists used an analytical pipeline to quantify the web-making

repeated the steps programmatically to create a Markov model (chain in which state transitions depend on the current state and an action vector).

This led to the discovery that web-building phases are defined by stereotyped sequences of actions that are mainly shared across individuals, according to a hierarchical hidden Markov model. The model comes into effect regardless of whether these stages develop in a normal or atypical manner. Action sequences alone might predict web stages, indicating that web-stage geometries are a physical expression of behavioural and instinctive transition regimes.

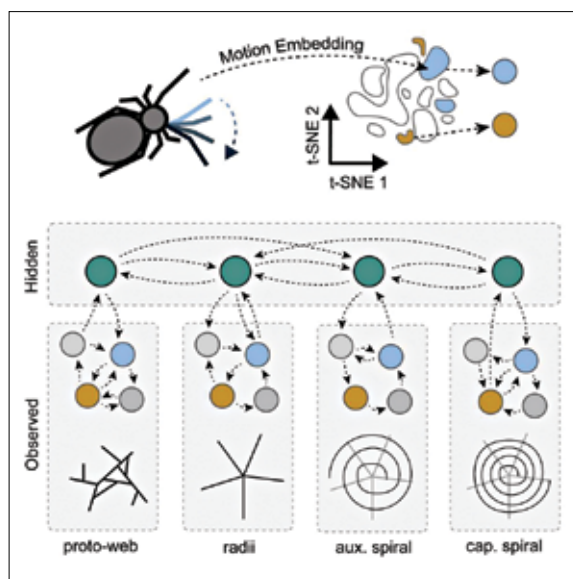
SPIDER, THE PREDATOR

When a spider catches prey in its web's sticky strands, it approaches the bug and injects venom with its fangs. The prey is either killed or paralysed by the

venom. Many times, when a prey is being eaten by the spider, it is not even completely dead. However, not all spiders feed on webs. Some spiders don't make any webs at all. Other spiders are on the hunt for their victim on foot. When their prey gets close, they build sticky nets to throw over it.

Many scientists are studying spider behaviour and spider silk in the hope of being able to harvest the material or, perhaps, genetically

alter it. This diverse silk could be used to improve the strength of body armour or generate skin grafts, for example. A few companies are currently investing in spider silk, including Kraig Biocraft Laboratories in Ann Arbor, Michigan, Spiber Technologies in Sweden, and AMSilk in Germany, which claims to have genetically engineered a protein that is similar to spider silk and is currently used in shampoos and other cosmetics.



behaviour of the orb-weaver in order to visualise the patterns in a systematic method. Several stereotyped stages of construction that could occur in regular or atypical progressions were revealed across individuals as a result of this pipeline and position tracking.

The scientists discovered general and stage-specific leg motions using an unsupervised clustering approach. As one leg motion led to the next typical stage of web-building, researchers